Analyzing Controversies in Political News Articles

Kseniia Ostudina **Raphael Leonardi** **Baptiste Pras**
Martin Leiva **Vladimir Herrera Nativi**

Université Paris-Saclay
{kseniia.ostudina, raphael.leonardi, baptiste.pras}@universite-paris-saclay.fr
{martin.leiva, vladimir.herrera-nativi}@universite-paris-saclay.fr

Abstract

Controversy detection asks whether a text concerns an issue on which the public is divided. Recent work simulates an audience of language model agents and measures their disagreement, but it targets social media, and few studies check how much their conclusions depend on the labels and on the prompt. We build a gold set of 100 political news articles, drawn with a stratified design from a Reddit-linked pool and labeled by five annotators, and we compare a six-persona audience simulation with a single language model agent, an aggregator agent and four supervised baselines. All systems run on one laptop with 7B and 9B models. The audience simulation never beats a single agent: on Qwen2.5 it is close to random (PR-AUC 0.26 against 0.57 for the single agent), and on GLM-4 it is not better. We find two reasons: the persona stance measures agreement with the article rather than a divided public, and six votes give a coarse score. The measurement itself is unstable. Changing the label source from humans to a language model annotator changes the winner (Kendall's $\tau = 0.56$), the Reddit and Wikipedia signal used for sampling agrees with humans at chance level, and prompt paraphrase moves scores as much as the gaps between systems. Code, labels and article identifiers are available at https://github.com/baptistepras/Analyzing-Controversies-in-Political-News-Articles.

1 Introduction

Some news topics divide the public into opposing camps, and others do not. Detecting the first kind automatically could help readers, moderators and researchers to see where a debate is happening. Early methods used structural signals such as Wikipedia edit wars or the shape of a discussion graph (Dori-Hacohen and Allan, 2015; Garimella et al., 2018). Recent methods use large language models (LLMs). One promising idea is to simulate an audience: several LLM agents with different profiles react to a piece of content, and the disagreement among them is read as controversy (Ding et al., 2026). This idea fits news articles well, because a new article has no comment section yet.

Two problems make results here hard to trust. First, labels come from very different sources, from engagement signals to LLM annotators to experts, and these sources do not encode the same notion of controversy. Second, LLM systems are sensitive to prompt wording and sampling, and papers rarely report how much their scores move. A difference between two systems means little if it is smaller than the change caused by a new prompt.

In this work we transfer audience simulation to English political news and we evaluate it with these two problems in mind. We ask two research questions. **RQ1**: does the ranking of systems survive a change of label source? **RQ2**: how large is the variation caused by prompt paraphrase and decoding seed, compared with the differences between systems? Our contributions are the following.

- A gold set of 100 political news articles, drawn from a Reddit-linked pool of misinfogeneral (Verhoeven et al., 2024) with a stratified design and known inclusion probabilities, and labeled by five annotators with an overlap subset for agreement.
- A comparison of four LLM systems (single agent, six-persona audience, aggregator, and a homogeneous ablation) on two backbones with four supervised baselines, under two label regimes, with 9 runs per LLM system and bootstrap intervals over articles.
- A clear negative result: the audience simulation does not beat a single agent, and we show with concrete diagnostics why it fails.
- Evidence that the label source and the prompt wording change conclusions as much as the choice of system does.

2 Related Work

Early work on controversy detection focused on Wikipedia and general web pages, using edit histories and mutual reverts as signals of disagreement (Sepehri Rad and Barbosa, 2012; Dori-Hacohen and Allan, 2013). Dori-Hacohen and Allan (2015) later removed the need for manual labels by mapping web pages onto Wikipedia articles with known controversy scores. On the news side, Choi et al. (2010) used hand crafted linguistic features to find controversial issues in news text, and Mejova et al. (2014) built a crowdsourced lexicon of controversial terms, showing that controversial coverage in U.S. news outlets uses more negative and biased language. Garimella et al. (2018) proposed a graph based method that models the discussion around a topic as a network and computes a controversy score from its structure. This structural approach became a common baseline in later work.

Later studies combined text and structure. Beelen et al. (2017) detected controversial news articles from the linguistic and emotional content of their comment sections. Kim and Allan (2019) proposed an unsupervised method that also generates short explanations for why an article is controversial. Sznajder et al. (2019) showed that the textual context around a concept is a strong signal of controversiality, using simple and language independent features. Benslimane et al. (2023) combined text embeddings with graph neural networks and found that fusing both signals outperforms text only or graph only models.

A closely related line of work targets media bias rather than controversy directly. Card et al. (2015) released the Media Frames Corpus, which annotates news articles with a fixed set of policy frames. Fan et al. (2019) introduced BASIL, a dataset of news article triplets annotated with lexical and informational bias spans, and found that informational bias is more frequent than lexical bias. Spinde et al. (2021) released BABE, a larger set of expert annotated sentences, and showed that fine tuned transformer models clearly outperform feature based classifiers. Krieger et al. (2022) further adapted RoBERTa to the bias detection domain and reached a higher F1 score on BABE. Horych et al. (2024) pretrained a single model on a large mixture of bias related tasks and obtained the best reported result on BABE, showing that multi task pretraining transfers well across bias detection tasks. Wessel et al. (2023) unified many of these datasets into a single benchmark, MBIB, to allow fair comparison across tasks.

Other related tasks include hyperpartisan news detection and political stance detection. Kulkarni et al. (2018) classified the political ideology of news articles using multiple views of the text. Kiesel et al. (2019) organized a shared task on hyperpartisan news detection with a large number of participating systems. Mohammad et al. (2016) and Li et al. (2021) introduced stance detection datasets built from tweets discussing controversial political targets.

More recent work applies large language models to controversy related tasks. Sun et al. (2023) introduced DELPHI, a dataset of controversial questions with human labels, and used it to evaluate how well large language models acknowledge and answer controversial questions. Arun et al. (2025) studied controversy detection under realistic class imbalance on political discussions from Reddit. Ding et al. (2026) proposed a multi agent framework built on large language models for controversy detection in social media videos, reaching the best reported results on a large scale benchmark. However, most large language model based methods target social media rather than news articles, which leaves a clear gap for methods that focus specifically on controversy in political news text.

Beyond this gap in coverage, the labels used across this literature come from very heterogeneous procedures, from Wikipedia metadata used as distant supervision (Dori-Hacohen and Allan, 2015) to expensive expert annotation (Spinde et al., 2021; Fan et al., 2019), and the reported scores are obtained on corpora with different definitions and different class balances (Wessel et al., 2023). How far conclusions about controversy detection depend on those choices has not, to our knowledge, been measured. We make that dependence our main object of study.

3 Methodology

3.1 Task and Research Questions

Given an article a, a system must output a continuous controversy score $C(a) \in [0, 1]$ expressing the degree to which the topic of a divides the public into opposing camps. Controversy is a property of the topic and its reception rather than of the wording of one text: a neutrally written article on a contested issue is controversial, while a strongly worded article on a settled one is biased but not

controversial. We require a continuous score so that systems can be compared by ranking and calibration under class imbalance.

Our two research questions concern whether measurements in this area can be trusted, rather than whether one more architecture wins.

RQ1 *Label provenance.* Labels from human annotators, from an LLM annotator, and from engagement and encyclopedic signals encode different notions of controversy. Does the relative ranking of systems survive a change of label source?

RQ2 *Measurement stability.* How large is the variation caused by prompt paraphrase and decoding seed, relative to the between-system differences such experiments are used to establish?

3.2 Audience Simulation

We transfer the audience-simulation framing of Ding et al. (2026) from short video to news text. The graph-based tradition (Garimella et al., 2018) measures controversy as the polarization of a real interaction graph, which does not exist yet when an article is published. We therefore simulate the readers. Rather than asking a model whether an article *is* controversial, we ask several reader agents how they *react* to it and measure their divergence.

We use $N = 6$ personas, each given as a system prompt with an age, a place, a job, a political position and a media diet (Appendix A): far left, center left, centrist, center right, far right, and a person who does not follow politics. Race and ethnicity are not part of the profiles. Each agent reads the title and lead of a and returns a one-sentence reaction and a stance $s_i \in \{-1, 0, +1\}$ towards the article's principal claim (-1 disagrees or is angered, 0 is neutral, $+1$ agrees or is reassured). Writing $\hat{\pi}$ for the empirical distribution of the six stances, the **audience** score is $C(a) = H(\hat{\pi})/\log 3$, the normalized entropy of simulated reader opinion: unanimity gives $C(a) = 0$ and a maximally split audience gives $C(a) = 1$.

In the **aggregator** variant, a judge agent receives the annotation guideline, the title, and the six reactions with their stances, and returns a score in $[0, 1]$ with a short summary of where the readers disagree. It isolates the contribution of the entropy aggregation itself. The N agents are N sequential calls to a *single* resident model, so adding agents costs inference time but not memory.

System	Params	Trained on	Peak mem.
TF–IDF + LR	—	our labels (CV)	<1 GB
RoBERTa-base (FT)	125M	our labels (CV)	8.7 GB
DA-RoBERTa (FT)	125M	our labels (CV)	10.8 GB
DA-RoBERTa (BABE)	125M	BABE only	10.8 GB
Qwen2.5 agents	7B, 4-bit	none	4.7 GB
GLM-4 agents	9B, 4-bit	none	5.3 GB

Table 1: Compared systems. All run on one laptop (Apple M5, 24 GB unified memory). Encoder peaks include training. The LLM systems share one resident model per backbone, so the number of agents changes runtime, not memory.

3.3 Compared Systems

The **single agent** receives the guideline, the title and the lead, and returns a score in $[0, 1]$ with a one-sentence reason. It is the natural reference: if simulating six readers is useful, the audience should beat one model asked directly. The **homogeneous ablation** replaces the six personas with six samples of the centrist persona, and is scored with the same entropy. It tests whether persona diversity adds anything over repeated sampling of one agent.

We use two 4-bit backbones, Qwen2.5-7B-Instruct (Qwen Team, 2024) and GLM-4-9B (Team GLM, 2024), both used by Ding et al. (2026). We report both rather than selecting one. Four supervised encoders complete the comparison (Table 1): TF-IDF with logistic regression, RoBERTa-base (Liu et al., 2019), and the domain-adapted RoBERTa of Krieger et al. (2022), all trained on our labels by cross-validation, plus the same domain-adapted model fine-tuned on BABE (Spinde et al., 2021) and used as is, with its probability of "biased" as the controversy score. The last one tests whether a bias detector transfers to controversy without any training on our labels.

4 Experimental Setup

Corpus and candidate pool. No public dataset provides controversy labels for English political *news articles*, so we built one. Articles come from misinfo-general (Verhoeven et al., 2024), about 4.2 million news articles from 2017 to 2022 with publisher-level bias and reliability labels from Media Bias/Fact Check.[1] We kept political articles with a topic-level rule: a BERTopic topic (Grootendorst, 2022) of the dataset is kept when at least 30% of its titles match a political lexicon (15% if one of its keywords is political). We excluded satire, kept

[1] https://mediabiasfactcheck.com

articles of 150 to 2,000 words, and removed exact and near-duplicate titles. This leaves 1,801,181 candidates from 432 publishers. We then kept only articles whose URL was posted on r/politics, r/news, r/worldnews or r/Conservative within 30 days of publication, using the Arctic Shift archive.[2] This gives a pool of 190,059 articles (10.6%). Year 2018 has no URLs in the dataset and is absent.

Proxy flag and sampling. Controversy is rare in random news, so we computed cheap signals before any annotation. Reddit rules fire on a high comment-to-score ratio (at least 2, with at least 20 comments), and, from 2020 on, when upvote ratios are available, on an upvote ratio of at most 0.80 or on upvote ratios that differ by at least 0.20 between subreddits. A Wikipedia rule fires when the title or the topic keywords match an issue page of Wikipedia's list of controversial issues.[3] An article is flagged H if any rule fires and L otherwise (28% of candidates are H). We drew 100 articles in two waves of 50. Each wave has 10 articles drawn uniformly at random from the pool, to estimate prevalence, and 40 stratified articles, balanced over the flag, the year, the publisher's political leaning (left, center, right) and reliability. We capped each publisher at 5 articles and each topic, month and publisher combination at 2. After wave 1, the share of H in the stratified part was lowered from 50% to 43%, because L articles had produced more of the minority class. Every article keeps its inclusion probability, so weighted estimates (Horvitz and Thompson, 1952) describe the pool. The final sample covers 50 publishers and 89 topics; 41% of its articles come from unreliable publishers, against 11% in the pool.

Gold labels. The five authors labeled every article with the question: does the passage discuss a topic that generates widespread moral, political or ideological debate, with sharply opposing camps? The guideline lists typical YES cases (military strikes, highly polarized legislation, debates with no consensus) and NO cases (accepted facts, routine announcements, shared humanitarian concerns), and states that a neutral tone does not change a divisive topic into a NO. Each label comes with a confidence (low, medium, high) and a one-line reason. Annotators saw only the title and the first 150 words, in random order, without the source, the stratum or the other labels. Twenty-six articles (13 per wave) were labeled by all five annotators, and the other 74 by one annotator each, balanced over strata. The gold label is the majority vote.

Silver labels and proxy labels. Silver labels come from Mistral-NeMo 12B,[4] a model family different from both backbones, given the full guideline, the title and the lead (temperature 0, one pass). For RQ1 we also read the proxy flag as a label (H as YES), to test the engagement and encyclopedic signal used for sampling.

Protocol. Each LLM system runs 9 times on all 100 articles: 3 paraphrases of every prompt (persona instruction, single agent, aggregator) times 3 decoding seeds. Persona and ablation calls use temperature 0.6, judge calls 0.3. Unreadable answers are retried up to twice with a new seed. The trained baselines use stratified 5-fold cross-validation repeated with 3 seeds, on the title and lead (256 tokens). Our main metric is PR-AUC with the minority class NO as the positive class (Saito and Rehmsmeier, 2015); the score of a system is inverted for this purpose. A random scorer gets the NO share, 0.24 on gold and 0.09 on silver. We also report precision at 10, macro-F1 at a 0.5 threshold, and expected calibration error (Guo et al., 2017) with 15 bins. Each score is the mean over runs with its standard deviation. For RQ2 we split the run-to-run spread into a prompt part (standard deviation over paraphrases of the seed-averaged score) and a seed part (the reverse). Because the gold set has only 24 NO articles, we also compute 95% intervals with a bootstrap over articles (2,000 resamples, the metric averaged over runs inside each resample), and paired intervals for differences between systems. For RQ1 we compare system rankings across label regimes with Kendall's τ (Kendall, 1938). Agreement is measured with Krippendorff's α (Krippendorff, 2004), Cohen's κ (Cohen, 1960) and Gwet's AC1 (Gwet, 2008). Qwen runs through MLX[5] and GLM through Ollama.[6] We report wall-clock time per 100 articles as the sum of call latencies.

[2] https://github.com/ArthurHeitmann/arctic_shift

[3] https://en.wikipedia.org/wiki/Wikipedia:List_of_controversial_issues

[4] https://mistral.ai/news/mistral-nemo

[5] https://github.com/ml-explore/mlx

[6] https://github.com/ollama/ollama

Label source	YES / NO	Agree	κ
Gold (majority of 5 humans)	76 / 24	—	—
Silver (Mistral-NeMo 12B)	91 / 9	81%	0.34
Proxy flag (H as YES)	42 / 58	48%	0.04

Table 2: Label sources on the 100 articles. Agreement and Cohen's κ are computed against the gold labels.

5 Experiments and Results

5.1 Annotation and Label Sources

The five annotators produced 204 labels. The gold set has 76 YES and 24 NO articles. Among the 20 articles drawn at random, 75% are YES (71% when all 100 articles are weighted by their inclusion probabilities), so most political news shared on these subreddits concerns a divisive topic under our guideline.

Agreement looks high but hides an asymmetry (Table 2). On the 26 articles labeled by all five annotators, 18 are unanimous, and all 18 are YES. No article is a unanimous NO: the three NO articles of this subset were all decided 3 against 2. Krippendorff's α is 0.26 (95% bootstrap interval 0.09 to 0.40), while pairwise raw agreement ranges from 73% to 92% and Gwet's AC1 from 0.65 to 0.90. This gap comes from the strong imbalance between YES and NO, which lowers chance-corrected scores even when raw agreement is high, but here it also points at a real difficulty: annotators reported low confidence on 22% of their NO labels and on 6% of their YES labels, and 21 of the 24 gold NO articles rest on a single annotator. The minority class, which is our positive class, is also the least reliable one.

The silver annotator says NO only 9 times and agrees with the gold labels on 81% of the articles ($\kappa = 0.34$). It finds 7 of the 24 gold NO articles. On the 26 articles with five human labels, it matches the human majority on 25. The proxy flag behaves very differently: read as a label, it agrees with the gold labels on 48% of the articles, with $\kappa = 0.04$. NO articles are not rarer among H articles (21%) than among L articles (26%). The engagement and encyclopedic signal we used to enrich the sample does not predict what humans call controversial. This also explains why the sample has 24% NO rather than the 30% we aimed for: the shift towards L articles in wave 2 could not help.

5.2 Main Comparison

Table 3 and Figure 1 report all systems. Under gold labels, the best LLM system is the single agent on Qwen (PR-AUC 0.566), far above random (0.24). The audience on the same backbone reaches 0.263, barely above random, and it is worse than the single agent in all 9 runs. On GLM, the audience reaches 0.399 against 0.454 for the single agent, better in 2 of 9 runs. The aggregator improves on the raw entropy in 9 of 9 runs on Qwen and 7 of 9 on GLM, but it does not beat the single agent on either backbone: on GLM the two are equal (0.450 against 0.454, better in 3 of 9 runs).

The best score overall comes from the domain-adapted RoBERTa trained on BABE and used as is (0.665), which needs about six seconds for 100 articles. It is ahead of the same model fine-tuned on our labels (0.499). Among the trained baselines, the domain-adapted model is ahead of TF-IDF (0.416) and RoBERTa-base (0.315).

With only 24 NO articles, the intervals are wide (Figure 1), so we test differences with a paired bootstrap (Table 4). On gold labels, three differences exclude zero: the Qwen single agent beats the Qwen audience and the Qwen aggregator, and the BABE model used as is beats the fine-tuned one. The other differences stay inside the noise, including the GLM single agent against the GLM audience and aggregator, and the best LLM system against the BABE model and against TF-IDF. On silver labels, the single agent beats every audience variant on both backbones. We therefore conclude that the audience simulation does not help, but we cannot claim that LLM systems beat simple supervised classifiers on this data.

5.3 RQ1: Label Provenance

The ranking of the ten main systems changes when the labels change (Table 3). Kendall's τ between the gold and silver rankings is 0.56. The bottom of the ranking is stable: the Qwen audience is last under both regimes, and the audience variants stay below the single agents. The top is not. The BABE model drops from first to fourth, and the GLM single agent rises from fourth to first. A plausible reason is that the silver annotator is itself an LLM that reads the same guideline, so it rewards the systems that judge like it. Its agreement with the human majority on the five-annotator subset (25 of 26) is higher than its agreement with single annotators, which shows that silver labels are not

System	Backbone	Gold PR-AUC mean ± std (rank)	Gold PR-AUC 95% CI	Silver PR-AUC (rank)	P@10	F1	ECE	Calls	Min/100
Random scorer	—	0.240	—	0.090	0.24	—	—	—	—
TF-IDF + LR	—	0.416 ± 0.050 (6)	[0.27, 0.59]	0.368 ± 0.032 (5)	0.43	0.55	0.20	—	0.0
RoBERTa-base FT	—	0.315 ± 0.017 (9)	[0.22, 0.45]	0.212 ± 0.097 (7)	0.27	0.53	0.24	—	4.2
DA-RoBERTa FT	—	0.499 ± 0.033 (3)	[0.35, 0.69]	0.423 ± 0.063 (3)	0.57	0.68	0.17	—	4.0
DA-RoBERTa (BABE)	—	**0.665** (1)	[0.49, 0.82]	0.406 (4)	0.70	0.58	0.33	—	0.1
Single agent	Qwen	0.566 ± 0.035 (2)	[0.40, 0.73]	0.458 ± 0.083 (2)	**0.74**	**0.72**	0.11	1	6.7
Audience	Qwen	0.263 ± 0.045 (10)	[0.18, 0.39]	0.104 ± 0.030 (10)	0.28	0.51	0.23	6	28.8
Aggregator	Qwen	0.372 ± 0.069 (8)	[0.26, 0.51]	0.252 ± 0.088 (6)	0.47	0.56	0.11	7	36.4
Homogeneous	Qwen	0.263 ± 0.040	—	0.111 ± 0.033	0.26	0.47	0.39	6	30.2
Single agent	GLM	0.454 ± 0.046 (4)	[0.30, 0.62]	**0.625** ± 0.138 (1)	0.67	0.63	0.15	1	8.9
Audience	GLM	0.399 ± 0.045 (7)	[0.28, 0.55]	0.144 ± 0.049 (9)	0.49	0.62	0.16	6	35.3
Aggregator	GLM	0.450 ± 0.064 (5)	[0.31, 0.59]	0.204 ± 0.069 (8)	0.62	0.69	0.25	7	45.0
Homogeneous	GLM	0.285 ± 0.024	—	0.097 ± 0.012	0.30	0.50	0.35	6	25.8

Table 3: Main results. PR-AUC uses NO as the positive class (gold: 24 NO of 100; silver: 9 NO of 100). Mean and standard deviation over 9 runs for LLM systems and 3 cross-validation repeats for trained baselines; DA-RoBERTa (BABE) has one deterministic run. The 95% interval is a bootstrap over articles. Ranks are among the 10 main systems. P@10, macro-F1 at threshold 0.5 and ECE are computed on gold labels. Min/100 is the time for 100 articles, as the sum of call latencies for LLM systems and as total cross-validation time for trained baselines.

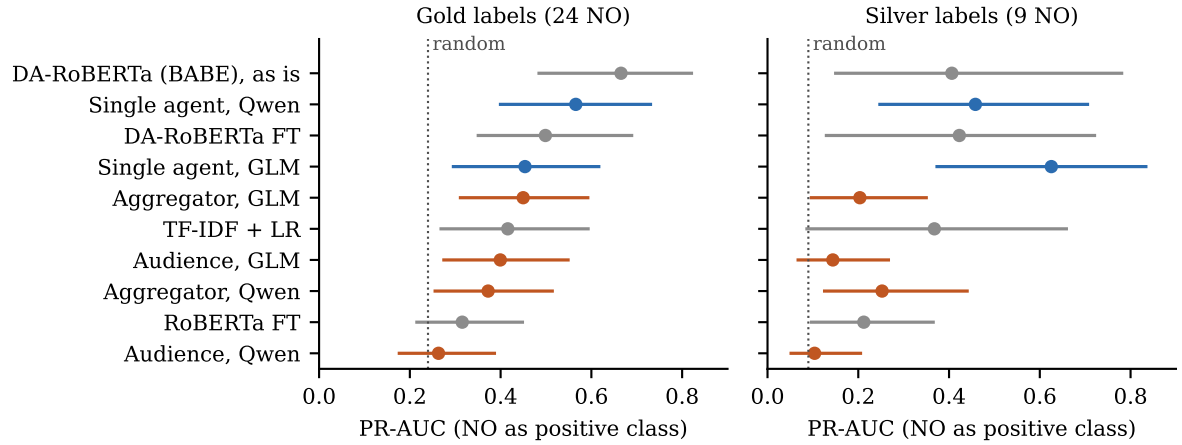


Figure 1: PR-AUC (NO as positive class) with 95% bootstrap intervals over articles. Grey: supervised baselines; blue: single agents; orange: audience and aggregator. The dotted line is a random scorer.

Comparison	Gold	Silver
Single − audience, Qwen	**+0.30** [+0.15, +0.44]	**+0.35** [+0.17, +0.57]
Single − aggregator, Qwen	**+0.19** [+0.06, +0.32]	**+0.21** [+0.01, +0.41]
Single − audience, GLM	+0.05 [−0.12, +0.22]	**+0.48** [+0.27, +0.67]
Single − aggregator, GLM	+0.00 [−0.14, +0.14]	**+0.42** [+0.23, +0.61]
Aggregator − audience, GLM	+0.05 [−0.02, +0.11]	**+0.06** [+0.01, +0.11]
Diverse − homog., GLM	**+0.11** [+0.04, +0.21]	**+0.05** [+0.01, +0.12]
Diverse − homog., Qwen	+0.00 [−0.06, +0.08]	−0.01
BABE − single, Qwen	+0.10 [−0.08, +0.28]	−0.05 [−0.35, +0.37]
BABE − DA-RoBERTa FT	**+0.17** [+0.09, +0.22]	−0.02 [−0.14, +0.18]
Single, Qwen − TF-IDF	+0.15 [−0.06, +0.32]	+0.09 [−0.20, +0.39]

Table 4: Paired differences in PR-AUC with 95% bootstrap intervals over articles. Bold: the interval excludes zero.

noise, but they are not a neutral substitute either. The third source, the proxy flag, is at chance level against the gold labels. A study that used it as its label would measure something unrelated to our human judgment. The answer to RQ1 is therefore that the choice of the winning system depends on who labeled the data, while the failure of the audience simulation does not.

5.4 RQ2: Measurement Stability

For most systems, the prompt wording moves the score more than the seed (Table 5). The effect is large for the single agent, which is the strongest LLM system. Averaged over seeds, the Qwen single agent scores 0.52, 0.58 and 0.60 on gold labels with the three paraphrases. On silver labels, the GLM single agent scores 0.55, 0.55 and 0.78, and the Qwen single agent 0.35, 0.53 and 0.50. These ranges (up to 0.09 on gold and 0.24 on silver) are as large as most gaps between systems in Table 3.

System		Gold		Silver	
		prompt	seed	prompt	seed
Single agent	Qwen	**0.037**	0.005	**0.093**	0.012
Single agent	GLM	**0.046**	0.009	**0.135**	0.054
Audience	Qwen	**0.043**	0.021	**0.023**	0.020
Audience	GLM	0.017	**0.024**	**0.023**	0.021
Aggregator	Qwen	**0.058**	0.015	**0.086**	0.029
Aggregator	GLM	0.017	**0.063**	0.013	**0.058**

Table 5: Run-to-run standard deviation of PR-AUC split into a prompt part (across 3 paraphrases) and a seed part (across 3 seeds). Bold: the larger part.

With a single prompt, the Qwen single agent would rank below or above the fine-tuned DA-RoBERTa on silver labels depending on the paraphrase. The GLM aggregator is the exception: its spread comes mostly from the seed, and its standard deviation over all runs (0.064) is the second largest on gold labels. Averaging the 9 runs per article helps: the per-article mean score of the GLM aggregator reaches a PR-AUC of 0.654 on gold labels, against 0.450 for single runs, and the Qwen single agent reaches 0.673. These ensemble numbers come at 9 times the cost and are exploratory, since part of the gain comes from breaking ties between the very discrete scores of single runs (Section 6).

5.5 Ablation and Cost

Persona diversity helps on GLM only. With six diverse personas, the GLM audience reaches 0.399, against 0.285 for six samples of the centrist persona. It is better in 9 of 9 runs, and the paired interval excludes zero on both label regimes (Table 4). On Qwen the two are equal (0.263 and 0.263, diverse better in 4 of 9 runs). Diversity creates disagreement on both backbones: the mean entropy rises from 0.42 to 0.70 on Qwen and from 0.43 to 0.69 on GLM. But on Qwen this extra disagreement carries no information about the gold label.

The audience costs 4.3 (Qwen) and 4.0 (GLM) times the single agent, and the aggregator 5.4 and 5.1 times. The full set of LLM runs took about 23 hours on one laptop, and the BABE model needs seconds. Under our results, the extra cost of simulating an audience buys no accuracy.

6 Discussion

Why the audience fails. Three diagnostics point to the same cause. First, the persona stance measures agreement with the article's claim, not whether the public is divided on the topic. When an article attacks a target that most readers dislike, all personas disagree with it and the entropy is low. For example, an article titled "Fauci funds absurd experiment to try to make male monkeys transgender" was labeled YES by all five annotators, but on Qwen 96% of all persona stances over the 9 runs rejected its claim, and its mean audience score is 0.09. The reverse also happens: "Joe Biden, Kamala Harris named Time's 2020 Person of the Year" was labeled NO, but the Qwen personas split along party lines and its mean audience score is 0.93. The single agent, which reads the guideline, scores these two articles 0.80 and 0.20. Second, the personas do not split along ideology as often as the design assumes. The far-left and far-right personas give the same stance on 51% (Qwen) and 54% (GLM) of the articles, and the center-left and center-right personas take opposite stances on only 11% and 23%. Third, the score is coarse: six votes on three values give only seven possible entropy values, and 42% to 46% of articles get an entropy of at least 0.9 whatever their topic. These three problems are properties of the design, not of one backbone, which suggests that a stance defined at the level of the topic ("would people unlike you disagree with you on this?") could fit the guideline better. We did not test it.

The backbones read the personas differently. The same persona behaves very differently on the two backbones (Figure 2). On Qwen the far-right persona disagrees with 65% of the articles, while on GLM it agrees with 73%. The two backbones give the same majority stance for this persona on only 32% of the articles. As a result, the per-article audience scores of the two backbones are almost unrelated (Spearman 0.19), while the single-agent scores agree well (0.76). Persona prompts do not control the simulated opinion as much as the framing suggests, a point already raised for LLM opinion simulation in general (Argyle et al., 2023; Santurkar et al., 2023).

What the aggregator does. The aggregator is better than the raw entropy, but its scores correlate with the single agent's scores (Spearman 0.55 on Qwen, 0.52 on GLM) more than with the entropy it reads (0.18 and 0.45). It receives the guideline and the title, so it largely reproduces the judgment of a single agent and uses the reactions as extra context. This explains why it can match the single agent on GLM but not beat it. Its scores are also very discrete: on GLM 51% of its answers are exactly

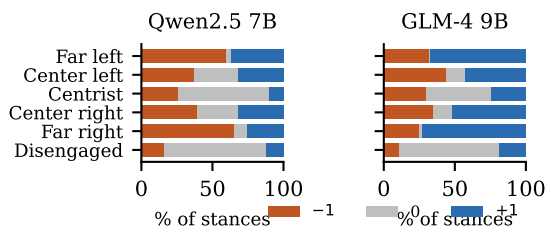


Figure 2: Share of stances -1, 0 and $+1$ per persona, over all runs and articles, for the two backbones.

0.5, and on Qwen 51% are exactly 0.8.

The bias detector. The strongest single number comes from a model trained to detect biased wording, used without any training on our labels. This seems to contradict our definition, in which wording does not decide controversy. We see two explanations. Our sample over-represents unreliable publishers (41%), where divisive topics and biased wording often occur together. And the result is fragile: the model has a single run, its interval overlaps those of the single agents, it is badly calibrated (ECE 0.33), and it drops to fourth place on silver labels.

Implications for evaluation. Our two research questions have practical answers. For RQ1, a leaderboard built on LLM labels favors LLM systems, and a leaderboard built on engagement and encyclopedic signals may not measure human judgment at all; both should be read next to human labels, not instead of them. For RQ2, a single prompt and a single seed would have produced a different leaderboard, and only the largest gaps in our study survive a bootstrap over articles. Reporting several paraphrases, the share of variance they cause, and intervals over items costs little compared with the risk of a wrong conclusion.

7 Conclusion

We transferred audience simulation from social media video to political news articles, built a 100-article gold set with a stratified, Reddit-linked sampling design, and compared four LLM systems on two small backbones with four supervised baselines. The audience simulation never beats a single LLM agent asked directly, at four to five times the cost. The main reason is a mismatch between what the personas are asked (do you agree with this article?) and what the task needs (is the public divided on this topic?). The measurement around this result is fragile: changing the label source changes the winning system, the proxy signal used for sampling agrees with humans at chance level, and prompt paraphrase moves scores as much as the gaps between systems. Future work should test topic-level stances, more personas, and larger backbones, and should grow the minority class so that smaller differences can be measured.

Limitations

The gold set is small: 100 articles with 24 NO, so only large differences can be detected, and most of our comparisons stay inside the bootstrap noise. Most NO labels come from a single annotator, and agreement on NO is low. The sample was drawn from articles shared on four large subreddits and is stratified, so the unweighted metrics describe this sample, not political news in general; year 2018 is absent, and the sample over-represents unreliable and centrist publishers by design. We used only two backbones of 7B and 9B parameters, three paraphrases and three seeds; larger models might simulate personas more faithfully. Of the three ablations we designed, only persona diversity was run. The silver labels come from a single LLM pass, and the proxy flag was built for sampling, not as a label. Trained baselines use cross-validation on the same 100 articles, which can favor them, although the sample covers 89 topics. Annotators and personas only saw the title and the first 150 words. Finally, latencies were measured on one laptop that was also used for other tasks, so they are orders of magnitude rather than precise benchmarks.

Ethics Statement

The articles come from misinfo-general, distributed under a CC BY-NC-SA license with a request not to redistribute the text openly; we release only article identifiers, labels and metadata, not article text. Reddit data were used only as aggregate signals (post counts, scores, upvote ratios) and no user information is stored. The personas describe political positions and demographic details of fictional people; we avoided race and ethnicity, but persona prompts can still produce stereotyped reactions, and our results show that they are unstable across models. A controversy score should not be used to hide or demote content, because it reflects a con-

tested judgment on which even our five annotators disagreed. The five annotators are the authors of this paper.

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A Personas and Prompts

Each persona is a system prompt ending with "Answer as this person genuinely would, not as an AI assistant." The six profiles are:

- **Far left**: a 24-year-old union organizer in Minneapolis, on the far left of the Democratic party, closer to Bernie Sanders than to the party establishment; reads Jacobin, The Intercept and political social media.
- **Center left**: a 45-year-old high school teacher in a suburb of Philadelphia, a mainstream Democratic voter; reads NPR and the New York Times.
- **Centrist**: a 41-year-old suburban parent with no strong party affiliation, who skims headlines once a day and rarely reads past the first paragraph.
- **Center right**: a 52-year-old car dealership manager in an exurban town in Ohio, a mainstream Republican voter not aligned with the populist wing; reads the Wall Street Journal and watches local TV news.
- **Far right**: a 58-year-old small-business owner in a rural US county, a firm Trump supporter, who gets news from conservative talk radio and local TV.
- **Disengaged**: a 29-year-old delivery driver in Atlanta with long, irregular hours, who does not follow politics or the news.

The first paraphrase of the persona instruction reads: “Read the article below the way the person you are would, then answer with ONLY a JSON object and nothing else, in exactly this form: {"reaction": "<one short sentence, in character>", "stance": <int>}. stance meaning: -1 disagrees with or is angered by the article’s principal claim, 0 is neutral or unmoved, 1 agrees with or is reassured by it.” It is followed by the title and the 150-word lead. The single agent and the aggregator receive the guideline (question, YES and NO definitions, indicators, and the two edge cases), and must return a JSON object with a score between 0 and 1 and a short reason or summary. The other two paraphrases reword the same content and change the order of the instructions.

B Implementation Details

Setting	Value
Qwen checkpoint	Qwen2.5-7B-Instruct, MLX 4-bit
GLM checkpoint	GLM-4-9B, Ollama 4-bit
Temperature	personas 0.6; single agent, aggregator 0.3
Max new tokens	personas 120; single agent, aggregator 200
Retries	up to 2, with a new seed
LLM runs	3 paraphrases $\times$ 3 seeds
Mean latency per call	Qwen 3.1 s, GLM 3.4 s
Encoders	4 epochs, batch 16, lr $2 \cdot 10^{-5}$, 256 tokens
TF-IDF + LR	1 to 2-grams, balanced class weights
Cross-validation	stratified, 5 folds $\times$ 3 seeds
Silver annotator	Mistral-NeMo 12B, temperature 0, 1 pass
Bootstrap	2,000 resamples of the 100 articles

Table 6: Hyperparameters and settings.

Out of 25,204 LLM calls, 4 GLM answers could not be parsed at first and were all recovered by a retry; no article was left without a score.